

Bulkly: Project Implementation Strategy

1. Product Evolution (Roadmap)

The project tracks the development of a bulk messaging utility through three primary phases:

- **Phase 1 (V1):** Initial automation focused on broadcasting copy-paste generated content.
- **Phase 2 (V2):** Introduced manual text entry with pre-configured timing delays to manage message pacing.
- **Phase 3 (V3):** Current active development phase focusing on system stability and expanded functionality.

2. Technical Debt & Critical Bug Fixes

To ensure the program is production-ready, the following architectural issues must be addressed during the initial coding phase:

- **Session Persistence:** Implement secure session storage to eliminate the requirement for users to re-authenticate with WhatsApp every time the program initializes.
- **Message Queue Logic:** Redesign the startup sequence to prevent the automatic, unprompted retry of failed messages.
- **Rate Limiting & Safety:** Develop a "Text Limiter" to control outbound volume and protect user accounts from being flagged for spam.
- **Execution Stability:** Optimize the background execution process to prevent sequence breaks caused by system sleep modes or timing inconsistencies.

3. Future Feature Backlog (Implementation Goals)

Once core stability is achieved, the following features will be integrated to enhance the "centralized unit" vision:

A. User Customization & Dynamic Templating

- **Editable Metadata:** Expand the configuration settings to allow users to modify organization names, physical addresses, and contact phone numbers within the message templates.
- **Thematic Generation:** Create a theme-selection engine where message generation aligns with specific user-defined styles or campaign objectives.

B. Intelligent Interaction

- **Bidirectional Communication:** Transition from a broadcast-only tool to a system capable of handling incoming replies.
- **Sentiment-Aware Responses:** Integrate a "Smart Reply" module to automatically respond to client inquiries while maintaining an understanding of the client's tone and mood.

C. Omnichannel & Social Integration

- **Cross-Platform Hub:** Extend API support beyond WhatsApp to include the Meta suite (Instagram/Facebook), TikTok, X (Twitter), and Telegram.
- **Batch Community Management:** Develop a utility to automate the addition of contacts to WhatsApp Groups or Channels in a single batch operation.

4. Recommended Development Priorities

1. **Stability First:** Solve the login persistence and sleep-mode sequence breaks before adding new features.
2. **Modular Design:** Build the "Future Vision" as separate modules (e.g., an AI-Reply module and a Platform-Connector module) to allow for easier omnichannel scaling.
3. **Variable-Based Templates:** Ensure the core message engine uses variables for organization details to meet the "Usable for All" requirement.